

**STATE OF CALIFORNIA
CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION
81 Higuera Street, Suite 200
San Luis Obispo, California 93401-5427**

WASTE DISCHARGE REQUIREMENTS ORDER NO. 00-043

Waste Discharger Identification No. 3 440107001

**FOR
SAN LORENZO VALLEY WATER DISTRICT,
BEAR CREEK ESTATES WASTEWATER TREATMENT FACILITY,
SANTA CRUZ COUNTY**

The California Regional Water Quality Control Board, Central Coast Region (hereinafter called the Board) finds that:

SITE OWNER AND LOCATION

1. On January 27, 2000, James A. Mueller, San Lorenzo Valley Water District (SLVWD) District Manager, filed a Report of Waste Discharge. The report was filed on behalf of Bear Creek Estates Wastewater Treatment Facility for authorization to continue discharging treated domestic wastewater within the San Lorenzo River sub-basin.
2. San Lorenzo Valley Water District (hereafter Discharger), 13060 Highway 9, Boulder Creek, owns and operates a wastewater treatment facility. The facility includes collection lines and a septic-tank/leachfield system. It is located in Santa Cruz County along Bear Creek Road approximately 2 1/2 miles northeast of the community of Boulder Creek as shown on Attachment "A" of this Order.

PURPOSE OF ORDER

3. The primary objectives of this order are to: 1) Permit the discharge of treated domestic wastewater; 2) Review and revise discharge limits; and 3) Review and revise a discharge monitoring program to evaluate the impact to water quality.

SITE/FACILITY DESCRIPTION

Discharge Type

4. The facility is designed to treat and dispose domestic wastewater from the Bear Creek Estates Subdivision, Units 3, 4, and 5.

Design and Current Capacity

5. The facility treatment system consists of two single-chambered septic tanks in series, each having a liquid volume of approximately 16,800 gallons (total capacity of approximately 33,600 gpd).
6. Septic tank effluent is alternately discharged among five leachfields.
7. The facility is designed to treat an average dry weather flow of 12,000 gallons per day gpd and a peak wet weather flow of 32,500 gpd.
8. The system treats and disposes up to 16,500 gallons-per-day (gpd) of domestic wastewater.

Geology

9. The leachfield disposal area is located on approximately 2.1 acres of sloping (5-30%) topography consisting of silty/clay soils underlain by siltstone and sandstone.

Surface and Groundwater

10. Bear Creek, a tributary to San Lorenzo River, flows around the eastern portion of the disposal area. The edge of the leachfield disposal area is over 100 feet from the creek.
11. Depth to ground water was measured during wet weather at a depth of 12 feet and generally migrates in a southerly direction. Ground water in the vicinity of the discharge has the following average concentrations based on results of ground water samples collected from March 1986 to September 1994:

Constituent	Concentration
Total dissolved solids	552 mg/l
Sodium	102 mg/l
Chloride	31 mg/l
Total nitrogen	3.4 mg/l
Nitrate as NO ₃	1.8 mg/l

Ground water samples collected from December 1991 to December 1999 have the following average concentrations:

Constituent	Concentration
Total dissolved solids	503 mg/l
Sodium	95 mg/l
Chloride	36 mg/l
Total nitrogen	7.4 mg/l
Nitrate as NO ₃	1.4 mg/l

12. Storm water flows over the leachfield area and into Bear Creek. Because the flow is less than one million gpd, the facility is not required to have a storm water pollution prevention plan.

Monitoring and Reporting Program (MRP)

The MRP requires effluent monitoring, ground water monitoring, receiving water monitoring, leachfield/seep monitoring, and system performance monitoring.

BASIN PLAN

13. The Water Quality Control Plan, Central Coastal Basin (Basin Plan), was adopted by the Board on November 17, 1989, and approved by

the State Water Resources Control Board on August 16, 1990. The Regional Board approved amendments of the Plan on February 11, 1994 and September 8, 1994. The Basin Plan incorporates statewide plans and policies by reference and contains a strategy for protecting beneficial uses of State waters. BP

14. Present and anticipated beneficial uses of Bear Creek that could be affected by the discharge include: BP

- Domestic and municipal supply;
- Agricultural supply;
- Ground water recharge;
- Industrial service supply;
- Water contact recreation;
- Non-contact water recreation;
- Wildlife habitat;
- Cold fresh-water habitat;
- Fish migration;
- Fish spawning
- Preservation of Biological Habitats of Special Significance;
- Rare, Threatened, or Endangered Species;
- Fresh Water Replenishment and,
- Commercial and Sport Fishing.

15. Present and anticipated uses of ground ^{water} in the vicinity of the discharge include: BP

- Domestic supply;
- Agricultural supply;
- Industrial process supply; and,
- Industrial service supply.

16. The San Lorenzo Wastewater Management Plan (WWMP), adopted by Santa Cruz County Board of Supervisors, was approved by the Regional Board on April 5, 1995 as Resolution 95-04. The WWMP includes findings and recommendations resulting from investigation of elevated nitrate levels in surface water and groundwater in the San Lorenzo River watershed. The WWMP recommends the Regional Board require nitrogen control measures in the issuance of new or revised waste discharge requirements. The WWMP's

goal is for at least 50 % reduction in nitrogen from onsite disposal systems. BP

17. This order requires 50 % reduction of nitrogen in effluent, consistent with Resolution No. 95-04. BP
18. The surface water quality objectives specified in the Basin Plan for the San Lorenzo River above Bear Creek, are: BP

Analyte	Value	Units
Total Dissolved Solids	400	mg/l
Sodium	50	mg/l
Chloride	60	mg/l
Boron	0.2	mg/l
Sulfate	80	mg/l

19. The range of median ground water quality objectives in the San Lorenzo Sub-Basin, as specified in the Basin Plan, are reported as follows: BP

Analyte	Range	Units
Total Dissolved Solids	100-250	mg/l
Sodium	10-20	mg/l
Chloride	20-30	mg/l
Nitrate (as N)	1-5	mg/l
Sulfate	10-50	mg/l

CEQA

20. These waste discharge requirements are for an existing facility and are exempt from provisions of the California Environmental Quality Act (Public Resources Code, Section 21000, et seq.) in accordance with Section 15301, Chapter 3, Title 14, of the California Code of Regulations.

EXISTING ORDERS AND GENERAL FINDINGS

21. The discharge has been regulated by Waste Discharge Requirements Order No. 95-09, adopted by the Board on March 10, 1995. The discharge from Bear Creek Estates has been regulated by the Board since February 15, 1963.
22. Discharge of waste is a privilege, not a right,

and authorization to discharge is conditional upon the discharge complying with provisions of Division 7 of the California Water Code and any more stringent effluent limitations necessary to implement water quality control plans, to protect beneficial uses, and to prevent nuisance. Compliance with this Order should assure this and mitigate any potential adverse changes in water quality due to the discharge.

23. On **March 28, 2000**, the Board notified the Discharger and interested agencies and persons of its intent to adopt waste discharge requirements for the discharge and has provided them with a copy of the proposed Order and an opportunity to submit written views and comments.
24. In a public hearing on July 14, 2000, the Regional Board heard and considered all comments pertaining to the discharge and found this Order consistent with the above findings.

IT IS HEREBY ORDERED, pursuant to authority in Section 13263 of the California Water Code, that San Lorenzo Valley Water District, its agents, successors, and assigns, may discharge wastewater at Bear Creek Estates Wastewater Treatment Facility providing compliance is maintained with the following:

[Note: Other prohibitions and conditions, definitions, and the method of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated January, 1984. Applicable paragraphs are referenced in paragraph E.3. of this Order.

Throughout the Order superscripts (BP) are provided to indicate requirements taken from the Basin Plan. Requirements not referenced are based on staff's professional judgment.]

PROHIBITIONS

1. Discharge of wastewater to the treatment system from other than Bear Creek Estates

Units 3, 4, and 5, is prohibited.

2. Discharge of wastewater is prohibited outside of the leachfield disposal area shown on Attachment "A".
3. Discharge of wastewater within 100 feet of Bear Creek is prohibited.
4. Wastewater on the soil surface of the leachfield disposal area is prohibited.
5. Bypass of the collection and treatment systems and discharge of untreated or partially treated wastes directly to the leachfield disposal area is prohibited.

SPECIFICATIONS

Effluent Limitations

1. Daily flow averaged over each month shall not exceed 16,500 gallons.
2. Effluent discharged to the leachfield shall not have a pH less than 6.5 or greater than 8.3BP.

Receiving Water Limitations

3. The discharge shall not cause a statistically significant increase in fecal coliform concentration in Bear Creek. Compliance is evaluated by statistically comparing fecal coliform concentrations in creek samples collected upstream and downstream of the leachfield disposal area.

Ground Water Limitations

4. The discharge shall not cause a statistically significant increase of mineral constituent (total dissolved solids, sodium, chloride, and nitrate) concentrations in the ground water. This is determined by statistical analysis of water quality data from the disposal area's monitoring wells.
5. The discharge shall not cause ground water to exceed concentrations of chemical constituents limits specified in Title 22, Chapter 15, Article

4, Section 64435, of the California Code of Regulations as listed below BP.

Constituents	Maximum Contaminant Level (mg/l)
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Nitrate (as NO ₃)	45
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System Operation Specifications

6. The wastewater discharge shall be rotated among the five leaching areas at least once every nine months.
7. High water alarms, readily audible or visible to maintenance personnel, shall be installed at each raw wastewater pumping station.
8. Stand-by power or portable pumps shall be available for use when wastewater-pumping facilities fail.
9. The Discharger currently has an onsite septic system. To comply with the San Lorenzo Wastewater Management Plan, this system shall be modified to reduce nitrogen levels by a minimum of 50%. The Discharger will comply with the timeline listed below for system construction:

Task	Completion Date
Completion of extended treatment system design	1/1/2002
Installation of extended treatment system	7/1/2003
Submit a technical report detailing optimal system performance	1/1/2004

Solids/Solid Waste Control

10. Solids accumulation in each septic tank shall be measured semi-annually and the appropriate tank cleaned when it appears (a) the bottom of the scum layer will be within 4" of the bottom of the outlet device before the next scheduled inspection or (b) the sludge level will be within 10" of the outlet device before the next scheduled inspection.
11. Solids cleaned from tanks shall be disposed of at an approved septage receiving facility.

Stormwater Control

12. Extraneous surface drainage shall be diverted away from the treatment system and leachfield disposal area.

Inflow/Infiltration Control

13. The Discharger shall develop and implement an Infiltration/Inflow and Spill Prevention Program (Program). The Program shall be reviewed and updated as necessary by every five years.
- A. The Program shall be developed in accordance with good engineering practices and shall address the following objectives:
 - a. identify infiltration and inflow sources that may affect treatment facility operation or possibly result in overflow or exceed pump station capacity; and,
 - b. identify, assign, and implement spill prevention measures and collection system management practices to ensure overflows and contribution of pollutants or incompatible wastes to Discharger's treatment system are minimized.
 - B. The District shall make a copy of the Program available upon request to a representative of the Regional Board.
 - C. The Program shall provide a description of the collection and transport system, measures used to ensure proper operation, and other

information necessary to determine compliance with these requirements. The Program shall include, at a minimum, the following items:

- a. A map showing: collection system lines greater than 12 inches, pump stations, standby power facilities, surface water bodies (including discharge point(s) where pump station overflows may occur), storm drain inlets, and date of last revision.
- b. A narrative description of the following:
 - i. Line Flushing and Cleaning: Describe available equipment and projected schedule necessary to clean and flush entire system every two years, and assigned staff. Describe coordination with area plumbers to address introduction of incompatible wastes (e.g., root balls, construction debris, etc.) during lateral cleaning and efforts to abate introduction of material into the system, which might cause system backup.
 - ii. Visual System Inspection: Describe visual inspection methods (e.g., televising lines), replacement schedules, frequency, collection system length and assigned staff. Describe results and detail problem areas found. Inspection records shall be retained for five years.
 - iii. Inflow & Infiltration: Describe current and five year projected investigation methods (e.g., smoke testing), frequency, results, and efforts to reduce storm water inflows and sewer line ex-filtration. Inspection records shall be retained for five years. Preventive Repair and Replacement: Describe a projected schedule to eliminate sewage conveyance systems determined or projected to be structurally compromised. List each project or reach of conveyance to be replaced separately along with proposed start and estimated completion dates.

- iv. Pump Station Maintenance: Describe each pump station, location, flow monitoring (wet and dry weather), and the previous year's operational problems and overflows.
 - v. Alternate Power Supply for Pump Station Operation: Describe alternate power supply for each pump station within the member entity's system.
- D. Fiscal Resources: The Program shall provide a description of fiscal resources necessary to ensure system operation. The Program shall include, at a minimum, the following items:
- a. Fee Structure: Quantification of current and five year projected sewer assessment fees necessary to implement the Program.
 - b. Available Fiscal Resources: Actual and five year projected budget expenses for staffing, operation and replacement of the collection system, including a description of a capital improvement or sinking fund to provide funding for item E.e., below.
- E. Personnel and Training: The Program shall provide a description of staffing available to ensure system operation. The Program shall include, at a minimum, the following items:
- a. Personnel: Identify specific individuals (and job titles) who are responsible for developing, implementing, and revising the Program. Provide an organizational chart of all staff, position, duties, and training received during the past year. Identify managers and provide a list of contacts with associated phone numbers.
 - b. Training: List the frequency of training, qualification of each employee. Periodic dates for training shall be identified.
- F. Planning and Reporting: The Program shall provide a description of planning efforts and reporting of system operation. The Program shall include, at a minimum, the following items:
- a. Spill Response: Describe a plan, identify employees responsible and duties necessary to implement your response to spills. Identify posting, notification, and spill estimation efforts used.
 - b. Annual Reporting: List spills or system problems during the previous year, cleanups, amounts, location, and efforts to ensure similar spills or problems do not reoccur. A tracking or follow-up procedure shall be used to ensure appropriate response has been taken. Inspections and maintenance activities shall be documented and recorded.
 - c. Offsite and Onsite Spill Alarms: Describe the current or proposed alarm system (or why unnecessary), central information location, staffing and response times for detecting spills from the system.
 - d. Wet Season Manhole Inspections: Describe or propose frequency to conduct inspections to detect line blockage during wet season flows to avoid system overflows, staffing, and available and projected equipment to ensure safe and effective inspections.
 - e. Capital Improvement - Describe a current and projected work plan.
 - f. Five Year Planning - Describe projected planning efforts.
 - g. 20 Year Planning - Describe long term planning efforts.
- G. The District shall provide an annual report, due January 30th of each year, describing program development and permit compliance over the previous twelve months. Report shall be of

sufficient content as to enable the Regional Board to determine compliance with requirements.

PROVISIONS

1. Order No. 95-09, "Waste Discharge Requirements for San Lorenzo Valley Water District, Bear Creek Estates, Santa Cruz County" adopted by the Board on March 10, 1995 is hereby rescinded.
2. Discharger shall comply with "Monitoring and Reporting Program No. 00-043", and any amendments thereto, as specified by the Executive Officer.
3. The Discharger shall comply with all items of the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated January, 1984; except Item Nos. A.8. and A11.
4. Pursuant to Title 23, Division 3, Chapter 9, of the California Code of Regulations, the Discharger must submit a written report to the Executive Officer not later than January 1, 2009, addressing:
 - a. Whether there will be changes in the continuity, character, location, or volume of the discharge; and,
 - b. Whether, in their opinion, there is any portion of the Order that is incorrect, obsolete, or otherwise in need of revision.

July 14, 2000

I, **Roger W. Briggs, Executive Officer**, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Coast Region, on July 14, 2000.

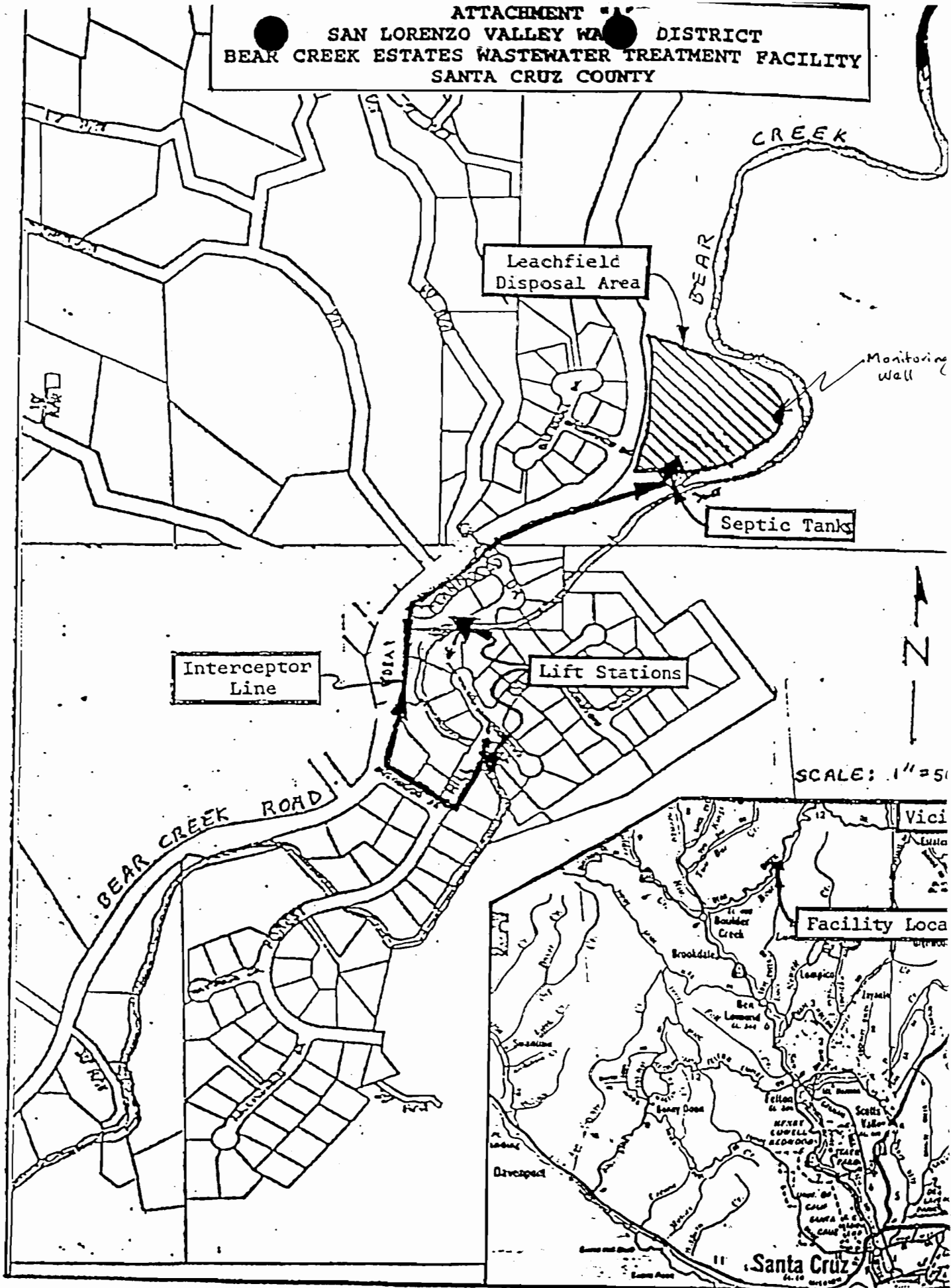
Bradley E. Hageman
for EXECUTIVE OFFICER

HEK/SLR/Permits/Bear Creek/00-043 WDR

Task: 121-01

File: Discharger file; San Lorenzo Valley WD, Bear Creek Estates WWF

ATTACHMENT
SAN LORENZO VALLEY WATER DISTRICT
BEAR CREEK ESTATES WASTEWATER TREATMENT FACILITY
SANTA CRUZ COUNTY



**STATE OF CALIFORNIA
CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION
81 Higuera Street, Suite 200
San Luis Obispo, CA 93401-5427**

**MONITORING AND REPORTING PROGRAM NO. 00-043
FOR
SAN LORENZO VALLEY WATER DISTRICT
BEAR CREEK ESTATES WASTEWATER TREATMENT FACILITY
SANTA CRUZ COUNTY**

INFLUENT MONITORING¹

Representative influent samples shall be collected before treatment and analyzed for the following:

Parameter	Units	Type of Sample	Minimum Measuring Frequency
BOD	mg/l	Grab	Monthly
pH	"	"	"
Organic Nitrogen (N)	"	"	* Twice per month
Nitrate (N)	"	"	"
Nitrite (N)	"	"	"
Ammonia (N)	"	"	"
Total dissolved solids	"	"	Quarterly
Total suspended solids	"	"	"

(N) – All nitrogen data shall be reported as nitrogen.

*Staff requires monitoring twice per month for the Nitrogen series (Organic Nitrogen, Nitrate, Nitrite, and Ammonia) for a minimum of 15 months. After 15 months Discharger shall submit a report analyzing the nitrogen removal efficiency of the wastewater treatment plant. Regional Board staff will evaluate the report to determine the need and frequency for continued nitrogen monitoring.

¹ Influent monitoring shall begin after installation of enhanced wastewater treatment system. The enhanced wastewater treatment system shall be installed by July 1, 2003.

EFFLUENT MONITORING

Representative effluent samples shall be collected prior to leachfield disposal and analyzed for the following:

Parameter	Units	Type of Sample	Minimum Measuring Frequency
Flow to Leachfield	Gallons per day	Metered	Monthly
BOD	mg/l	Grab	"
pH	"	"	"
Organic Nitrogen (N)	"	"	* Twice per month ²
Nitrate (N)	"	"	"
Nitrite (N)	"	"	"
Ammonia (N)	"	"	"
Total suspended solids	"	"	Quarterly
Total dissolved solids	"	"	"
Sodium	"	"	"
Chloride	"	"	"

(N) – All nitrogen data shall be reported as nitrogen.

*Staff requires monitoring twice per month for the Nitrogen series (Organic Nitrogen, Nitrate, Nitrite, and Ammonia) for a minimum of 15 months. After 15 months Discharger shall submit a report analyzing the nitrogen removal efficiency of the wastewater treatment plant. Regional Board staff will evaluate the report to determine the need and frequency for continued nitrogen monitoring.

² Effluent nitrogen monitoring shall be conducted quarterly until the installation of enhanced wastewater treatment system. The additional effluent nitrogen monitoring shall commence once the enhanced wastewater treatment system is installed or July 1, 2003, which ever comes first.

SEPTIC TANK MONITORING

Each septic tank shall be opened and measured for the following items:

Measurement	Units	Type of Measurement	Minimum Measuring Frequency
Distance of scum layer from bottom of outlet device	Inches	Staff gauge	Semi-Annually (June & Dec.)
Distance of sludge layer from bottom of outlet device	Inches	Staff gauge	Semi-Annually (June & Dec.)

Septic tanks shall be pumped as necessary to ensure proper performance and protection of the leachfield. In the event of a leachfield failure or impending failure, the Executive Officer and the County Environmental Health Department shall be notified immediately. In the event of failure, the area shall be posted to prevent public contact with domestic wastewater.

GROUND WATER MONITORING

The Discharger shall install one monitoring well upgradient and two wells downgradient of the leachfield disposal area. Prior to the installation of monitoring wells, Discharger shall submit a report to the Executive Officer discussing the proposed locations, monitoring well depths, and the technical justification of the proposal. The report shall include an estimated direction of groundwater flow and depth, to accurately position up gradient and down gradient wells. Reports detailing groundwater monitoring well installation and flow must be prepared by a registered professional engineer or hydrogeologist or equivalent professional knowledgeable in hydrogeology and ground water monitoring. The wells shall be installed by January 1, 2001.

The Discharger shall sample monitoring wells in accordance with the following schedule:

Constituent	Units	Type of Sample	Minimum Sampling and Analyzing Frequency
Total Dissolved Solids	mg/l	Grab	Semi-annually (June & Dec)
Sodium	"	"	"
Chloride	"	"	"
Organic Nitrogen (N)	"	"	"
Nitrate (N)	"	"	"
Nitrite (N)	"	"	"
Ammonia (N)	"	"	"
pH	"	"	"

BEAR CREEK MONITORING

Discharger shall establish two Bear Creek monitoring stations to determine impact of discharge. One station shall be located 100 feet upstream of the leachfield disposal area. The second station shall be located 100 feet downstream of the disposal area. Creek samples from these stations shall be collected and analyzed according to the following:

Constituent	Units	Type of Sample	Minimum Sampling and Analyzing Frequency
Fecal coliform	MPN/100 ml	Grab	Monthly*
Enterococci	"	"	"
Nitrate as N	mg/l	"	"
Total dissolved solids	"	"	"

*Monthly samples shall be collected during the 1st working day of the month. After one year of creek sampling (twelve samples) the Discharger may provide an analysis of coliform monitoring data to determine if a significant change has occurred due to the discharge. If no significant change has occurred, the Discharger may request a reduction in creek monitoring subject to Executive Officer approval.

DISPOSAL AREA INSPECTION

Monthly inspections shall be made of the leachfield disposal area for evidence of surfacing effluent. During the inspection, the level of effluent in each leachfield trench shall be recorded. Notes shall be kept of observations and shall be summarized in the Quarterly Monitoring Reports. In the event of a leachfield failure or impending failure, the Executive Officer and the County Environmental Health Department shall be notified immediately. In the event of failure, the area shall be posted to prevent public contact with domestic wastewater. Repairs shall be made as quickly as possible. To assure compliance with discharge specifications the discharger shall describe which leachfield is in use and when the leachfield was last rotated.

Monthly inspections shall be made of seep areas (along Bear Creek) for evidence of surfacing effluent. Notes shall be kept of observations and shall be summarized in the Quarterly Monitoring Reports. In the event of a surfacing effluent, the Executive Officer and the County Environmental Health Department shall be notified immediately. In the event of failure, the area shall be posted to prevent public contact with domestic wastewater. Repairs shall be made as quickly as possible.

REPORTING

Monthly sampling and inspection results shall be reported quarterly by the **1st day of February, May, August, and November**. Semi-annual sampling results shall be submitted with the January and July quarterly reports.

The quarterly reports shall also include a summary of sewage collection or treatment or disposal systems spill(s) during the reporting quarter. At a minimum, the following information shall be reported:

July 14, 2000

1. Location, volume, date, and time of sewage spill, and the number of times spill has occurred at this location within the last five years.
2. Corrective action(s) taken to stop and mitigate spill and prevent future spills.

If there is no spill during the reporting quarter, a "no sewage spill" statement shall be included in the quarterly report.

All data (weekly, monthly, and semi-annual) shall be submitted to the Board in both hard copy and electronic format. The electronic data submission shall conform to criteria approved by the Central Coast Regional Water Quality Control Board Executive Officer.

ORDERED BY Bradley E. Hagman
PE: Executive Officer

July 14, 2000
Date

HEK/SLR/Permits/Bear Creek/00-043 MRP

Task: 121-01

File: Discharger file; San Lorenzo Valley WD, Bear Creek Estates WWF